



Two-Stage Deep Ensemble Paradigm Based on Optimal Multi-scale Decomposition and Multi-factor Analysis for Stock Price Prediction

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Abstract

Stock price forecasting is important for financial risk management and investment decisions. However, traditional forecasting techniques are challenged and under pressure due to the complex characteristics of stock prices and the impact of quantitative trading. Therefore, to address these issues and produce a more accurate stock price forecasting method, this study proposes a two-stage deep integration paradigm based on optimal multi-scale decomposition and multi-factor analysis. This paradigm will also serve as a scientific support and reference for investors' actual investment decisions. (1) Optimal multi-scale decomposition methods are proposed to achieve decomposition of the closing price. Noise suppression and optimal decomposition modes are achieved by singular spectrum analysis (SSA) and variable mode decomposition (VMD) to reconstruct the high- and low-frequency sub-series. (2) The approximate entropy is used to measure the complexity of the time series, and the approximate entropy of the decomposed sequence is judged to reconstruct the characteristic subsequence. (3) The multi-factor analysis method uses a joint feature selection technique, the spearman correlation coefficient test, and the mutual information (MI) index to find the optimal feature influence factor and extract the optimal influence factor using feature compression. (4) Two-stage deep integration was performed using bidirectional gate recurrent unit (BIGRU) to obtain the final predictions. The two major indices of the Chinese A-share market, namely the Shenzhen Stock Index (SZI) and Shanghai Stock Exchange (SSEC), with data sourced from the Flush Finance website, were used to verify the validity of the model proposed in this paper. The MAE, RMSE, and MAPE values of the proposed model in this paper are 171.1673, 221.2086, and 1.2796% in SZI market and 29.9973, 39.3508, and 0.8946% in SSEC market, respectively, which are significantly better than other comparative models. Based on the experiments, it can be seen that the optimal multi-scale decomposition method performs better in capturing the hidden information of the original data, while the multi-factor analysis provides more relevant data for the prediction, which helps to improve the accuracy of the prediction. After integrating the predictions, the decomposition-integrated hybrid model proposed in this paper predicts significantly better than with other prediction models.

Keywords Multi-scale decomposition · Multi-factor analysis · BIGRU · Stock price forecast

Introduction

Background and Challenges

The stock price forecast is deeply concerned by investors and the financial industry. The prediction of stock price can not only avoid financial risks to a certain extent and improve the efficiency of financial market management [1], but also help

investors gain profits [2]. However, the stock price is vulnerable to many factors, such as its inherent randomness [3], exchange rate changes [4], and investor sentiment [5]. Therefore, finding trustworthy forecasting models to increase the accuracy of stock price projections is still a very difficult task. In order to help financial institutions more effectively deliver financial early warnings and assist investors in minimizing losses in the financial markets, highly precise stock price forecasting is crucial.

There is a wealth of research on stock price forecasting, which currently consists of statistical models and machine learning models. Traditional statistical models are usually used for financial time series forecasting with strict statistical assumptions and are mainly used to deal with linear

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problems [6]. However, since stock prices are non-stationary, non-linear, and chaotic, dealing with stock prices with statistical models is usually out of reach. In contrast, machine learning, especially deep learning models with strong non-linear feature capturing and data processing capabilities, is well suited to handle stock price data [7]. The fluctuations of stock prices are usually very complex, and in order to get more accurate prediction results, decomposition integration methods have thus become popular. This method decomposes the whole learning task into multiple subtasks and uses simpler models for modeling in each subtask. Then, by integrating the results of the subtasks, the final prediction results are synthesized.

Existing models on decomposing the integrated learning paradigm yield more accurate prediction accuracy, but still have some limitations. First, stock price prediction based on the decomposition integration framework usually ignores the impact of noise on stock prices. When predicting stock prices, noisy data may increase the uncertainty and error of the prediction, making it more difficult for the model to predict. In addition, in the use of decomposition methods, the optimal number of modes of decomposition of VMD is often neglected [8]. Too many modal numbers may generate additional noise and modal repetitions, while too few may filter critical information, both of which can affect the final prediction results. Second, the decomposition integration paradigm focuses on the decomposition of a single variable and tends to ignore the influence of extraneous factors. Whereas stock data are often heavily influenced by economic indicators, market sentiment, and other factors, taking multiple factors into account in the decomposition integration model will significantly improve the prediction accuracy [9–11]. In addition, due to the excessive number of influencing factors, reasonable feature screening and dimensionality reduction techniques are necessary to avoid dimensionality catastrophe and high-dimensional redundancy and to reduce computational complexity. Third, existing prediction methods based on the decomposition integration paradigm usually choose to integrate the subsequence results directly by linear accumulation [12] before inputting them into the final prediction model, which tends to ignore the complex features among subsequences, resulting in the accumulation of errors and affecting the prediction accuracy of the model.

To overcome these existing limitations, a two-stage deep integration paradigm based on optimal multi-scale decomposition and multi-factor analysis is used to predict stock prices. Noise reduction and scientific determination of the optimal number of decomposition modes are achieved through optimal multi-scale decomposition. Dimensionality reduction of multivariate data is achieved through joint feature screening technique and principal component analysis. Finally, the robustness of the model is improved by reducing the prediction bias through non-linear integrated prediction.

Novel Contributions

Considering the above limitations, this study proposes a two-stage deep integration paradigm based on optimal multi-scale decomposition and multi-factor analysis to predict stock prices. In the optimal multi-scale decomposition section, noise suppression and optimal decomposition modes are used to determine the optimal number of decompositions and to reconstruct the high- and low-frequency subsequences to avoid introducing higher computational complexity and error accumulation into the post-decomposition integration work. With the use of this decomposition method, BIGRU is able to learn non-stationary and non-linear time series features adaptively, enhancing the precision of later stock price index forecasts. Multi-factor analysis is then performed to find the optimal feature influence factor by joint feature selection techniques and extract the optimal influence factor by feature compression. This section takes into account the complexity of financial markets and improves the accuracy of forecasting by analyzing exogenous variables. At the same time, as too many feature influencing factors would increase the complexity of the calculation and weaken the forecasting effect if fed directly into the forecasting model, this paper uses feature compression to deal with this problem. Finally, integrated forecasting is carried out by inputting the high- and low-frequency series and their compressed influencing factors into BIGRU separately and then integrating their forecasting results into BIGRU to make a final forecast of the stock closing price.

The following are the main contributions of this paper:

1. Optimal multi-scale decomposition algorithm based on feature extraction and reconstruction. The method achieves noise reduction processing of stock price data under the decomposition integration framework and scientifically determines the optimal number of decomposition modes of VMD through sample entropy, which improves the feature extraction capability and the prediction accuracy of the model.
2. Multi-factor analysis based on decomposition integration framework. Most of the previous researches have only performed decomposition integration forecasting for single variables, often ignoring the extrinsic influencing factors of stock prices in modeling. In contrast, this paper not only considers these influencing factors, but also combines them into the decomposition integration framework and proposes a decomposition integration learning paradigm based on multi-factor analysis to improve the accuracy of forecasting. Meanwhile, the best influencing factors are better identified and extracted by the joint feature screening technique and

feature compression technique to improve the modeling capability on high-dimensional data.

3. A two-stage deep non-linear integration paradigm is used. Non-linear integration of the subsequence modeling results into the final prediction model not only overcomes the problems such as differences and correlations among different subsequences, which have been neglected in the previous literature by integrating subsequences in a linear cumulative manner, but also improves the prediction accuracy of the model.
4. An improved decomposition integration framework coupling optimal multi-scale decomposition and multi-factor analysis is proposed. In this paper, optimal multi-scale decomposition, multi-factor analysis, and non-linear integration are used to improve the prediction accuracy of the model, and the validity of the model is verified by empirical results.

Paper Organization

The remaining sections of this article are organized as follows. The “[Literature Review](#)” section reviews the literature on share price forecasting. The research techniques employed in this work are discussed in the “[Related Methodology](#)” section. The model put forward in this study is introduced in the “[Proposed Model](#)” section. The experimental procedure and empirical findings are introduced in the “[Empirical Study](#)” section. This paper’s conclusion and future study directions are covered in the “[Conclusions](#)” part.

Literature Review

Statistical Models

Financial time series forecasting has been receiving a lot of attention in recent decades. Traditional statistical models including autoregressive (AR) [13], autoregressive integrated moving average (ARIMA) [14], and generalized autoregressive conditional heteroskedasticity (GARCH) [13], which focus on normal and linear assumptions, are mainly used to deal with linear problems and show good performance [15]. Furthermore, the GARCH model solves the problem posed by the constant time series variance assumption and has good performance in predicting stock volatility [14].

Machine Learning Models

Stock price data are more irregular and non-linear due to noise interference, and it is difficult for statistical models to handle such complex time series [16]. Instead, machine

learning techniques imitate human learning behavior and automatically learn patterns or new information from data without prior knowledge, employing non-linear functions to find the best parameters with any level of precision. Therefore, despite the inconsistencies and non-linearities in stock price data, machine learning techniques can produce findings with high fitting accuracy. The stock market is highly dynamic and uncertain due to numerous factors, while stock prices are non-stationary, non-linear, and chaotic. Therefore, in order to predict stock prices more accurately, it is often necessary to use complex machine learning models, such as deep learning models [17]. Deep learning models such as recurrent neural network (RNN) can be used to extract time patterns from sequential data [18], improving the accuracy of model prediction to a certain extent, but it has the problems of gradient disappearance and gradient explosion, and still has long-term time dependence. Therefore, long short-term memory (LSTM) and gated recursive unit (GRU) are proposed. LSTM [19, 20] and GRU [21] are not only able to deal with the problems of gradient vanishing and gradient explosion, but also able to deal with the complex time series problems with higher fitting accuracy and ideal results, and obtain better prediction accuracy. Compared with LSTM, GRU is simpler and more efficient, has the advantages of shortening training time and improving network performance [22], and shows good performance in stock price prediction [23]. In addition, GRU-based hybrid models have been shown to be a new data augmentation approach for stock index trend prediction, and the approach is able to overcome the overfitting problem [21].

Decomposing and Integrating Predictive Models

Previous studies have found that a single prediction model is susceptible to random factors and has limitations and instability, and its prediction accuracy is worse than that of a hybrid model [24]. Through data preprocessing, feature engineering and other methods can effectively improve the poor generalization ability and dimensional explosion faced by single prediction models, which can effectively improve the prediction accuracy [25]. Among them, the most representative method is the decomposition integration method [26]. Decomposition ensemble learning paradigm mainly decomposes a complex sequence into several relatively stable and simple subsequences through signal decomposition technology, and then modeling. In this way, it can properly extract features and reduce uncertainty, which is very important for stock price prediction. Mainstream signal decomposition methods mainly include SSA [27], empirical mode decomposition (EMD) [28], and VMD [29]. SSA is suitable for analyzing time series problems with underlying structure, which can divide the original series into noise, oscillatory components, and slowly changing trends. The

decomposed series is then reconstructed into a smoother time series, which not only preserves the main components of the original series, but also serves the purpose of noise reduction of the series well. In addition, one of the most classical signal decomposition methods is EMD, which is a data-driven adaptive algorithm with the ability of local feature expression. It can well process non-linear and non-stationary time series, and efficiently and clearly identify the embedded structure [30]. However, EMD has the fatal problem of modal aliasing, which can occur when there is a perturbation of an abnormal event in the signal. The study of stock price time series needs to take into account the effects of unexpected events. To eliminate modal confusion, the VMD algorithm [31] is proposed. VMD can not only solve the problems of EMD described above, but also decompose the original signal into multiple components with specific bandwidth in the spectrum domain, and can fuse the decomposed signal into the actual signal. Bisoi et al. [32] have proved by experiments that VMD is an effective feature selection method. Hybrid forecasting models for financial time series based on the decomposition integration framework became popular as research progressed [33–35]. Experiments have shown that VMD-based hybrid models outperform other comparative models in terms of forecasting accuracy [36]. However, the multimodal time series obtained by the modal decomposition approach raises another problem, which is that the uncertainty and fine-grained number of modes bring higher computational complexity and error accumulation to the integration work after decomposition. Therefore, it is still a topic worth debating how many deconstructed modes can be determined scientifically. Furthermore, the decomposition integration paradigm focuses on the decomposition of a single variable and tends to ignore the influence of extraneous factors. Taking a complicated financial market as an example, stock price indices are influenced by natural, social, political, and economic elements. According to Lin and Yan [37], a number of different variables, which are input-dependent, can explain variations in stock price indices. Stock price predictions may suffer if these outside factors are ignored.

In summary, these studies are based on deconstructing the price change information from the original time series using modal decomposition methods in a decomposition integration framework and forecasting with integrated forecasting methods. At the same time, they almost always achieve better forecasting results relative to the benchmark model. It is noteworthy that these papers seldom consider the impact of the noise problem in the raw data on subsequent studies, and also ignore the impact of extraneous factors on the accuracy of stock price forecasts. Therefore, in this study, not only is the time series to be predicted noise-reduced, but a more scientific approach is used to determine the number of layers of modal decomposition and to reconstruct patterns

with pixel data characteristics into a smaller number of sub-series. The influence of extraneous factors on the time series to be predicted is also considered. Finally, the final prediction of the stock price is obtained by non-linearly integrating the predictions of each component.

Related Methodology

This paper constructs a two-stage deep integration paradigm based on optimal multi-scale decomposition and multi-factor analysis, and proposes a model in which multi-scale decomposition, multi-factor analysis, and bidirectional gate recurrent unit (BIGRU) are blended, and this decomposition integration paradigm can increase the accuracy of stock price forecasting. This section introduces the basic principles of the method used in this paper.

Singular Spectrum Analysis (SSA) Algorithm

SSA is a method for processing non-linear time series data [38], which is suitable for analyzing time series problems with underlying structure and can well achieve the purpose of denoising the series. SSA mainly consists of four steps, namely embedding, decomposition, grouping, and reconstruction, to extract different component sequences for the purpose of denoising the time series. The four main steps are as follows:

Step 1: Embedding. A one-dimensional time series of finite length $[x_1, x_2, \dots, x_N]$ is the object of analysis in SSA, where N is the length of the series. In order to generate the trajectory matrix, we must first select an appropriate window length (L) that lagged the original time series. The typical values are $L < N/2$, $K = N - L + 1$, and the trajectory matrix X is the matrix of $L \times K$.

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_K \\ x_2 & x_3 & \cdots & x_{K+1} \\ \vdots & \vdots & & \vdots \\ x_L & x_{L+1} & \cdots & x_N \end{bmatrix}. \quad (1)$$

Step 2: Decomposition. On the trajectory matrix, a singular value decomposition (SVD) is carried out. Specifically, X is decomposed into the following shapes.

$$X = U\Sigma V^T \quad (2)$$

where $U \in R^{L \times L}$ is the left matrix; $\Sigma \in R^{L \times K}$ is the singular value, that is, there are values on the main diagonal and all other elements are zero; and $V \in R^{K \times K}$ is the right matrix; in addition, U, V are all unit orthogonal matrices satisfying $UU^T = I, VV^T = I$. Equation (2) can be written

in the form of a sum of column vectors multiplied by row vectors as follows.

$$X = \sum_{i=1}^r \sigma_i u_i v_i^T = X_1 + X_2 + \dots + X_r, \quad (3)$$

where r is the rank of the matrix X , and also the number of non-zero singular values.

Step 3: Eigentriple grouping. The purpose of grouping is to divide the matrix X into linearly uncorrelated submatrices.

$$X = X_1 + X_2 + \dots + X_m. \quad (4)$$

The process of Eq. (4) is as follows: first, the singular values are divided into m groups, denoted as $\{I_1, I_2, \dots, I_m\}$, and the corresponding submatrices of each group are the matrices of the corresponding singular values in Eq. (3) multiplied by the left singular vector and the right singular vector. Then, after dividing the original time series into multiple disjoint groups, the original data is then divided into multiple sequences according to the magnitude of importance in preparation for the next step of reconstruction.

Step 4: Reconstruction. The most important aspect of this step is the anti-diagonal average. Suppose a matrix $T \in R^{L \times K}$ is averaged over all its anti-diagonals, and the analysis shows that there are a total of $N = L + K - 1$ anti-diagonals, then the formula for averaging the anti-diagonals of a matrix T is:

$$t_k = \begin{cases} \frac{1}{k} \sum_{m=1}^k T_{m, k-m+1}, & 1 \leq k < L * = \min\{L, K\} \\ \frac{1}{L*} \sum_{m=1}^{L*} T_{m, k-m+1}, & L * \leq k \leq K * = \max\{L, K\} \\ \frac{1}{N-K+1} \sum_{k=K*+1}^{N-K*+1} T_{m, k-m+1}, & K * < k \leq N \end{cases} \quad (5)$$

For grouping $\{I_1, I_2, \dots, I_m\}$, the matrices of each group are processed by inverse diagonal averaging; corresponding all to a sequence $\hat{y}_i (i = 1, 2, \dots, m)$ of length $N = L + K - 1$, m will be the vector summed to obtain the reconstructed value of the original time series Y_N . When selecting the reconstructed sequence, it is necessary to select the component with a large contribution from the previous m according to the size of the singular value to reconstruct the original sequence, thus achieving the purpose of denoising.

Variational Mode Decomposition (VMD) Algorithm

The VMD [39] is a variational modal decomposition paradigm that is entirely non-recursive. Around the central pulsation ω_k , which was found by the decomposition, each mode μ_k needs to be extremely compact. The squared L^2

-parameter of the gradient is minimized under the condition that the sum of the modes equals the original signal, whereas VMD requires that the sum of the bandwidths of the modes be calculated by the H^1 -Gaussian smoothness of the decomposed signal. Hence, solving the constrained variational issue for a time series f is similar to solving the decomposition problem.

$$\min_{\{\mu_k\}, \{\omega_k\}} \sum_k \|\partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * \mu_k(t) \right] e^{-j\omega_k t} \|_2^2, \quad (6)$$

$$\text{s.t. } \sum_k \mu_k = f$$

where $\{\mu_k\} := \{\mu_1, \mu_2, \dots, \mu_K\}$ and $\{\omega_k\} := \{\omega_1, \omega_2, \dots, \omega_K\}$ are shorthand notations for the set of K modes and their central frequencies, respectively. δ is the Dirac distribution, t is the time script, and $j^2 = -1$ and $*$ are the convolution operators. To make the problem unconstrained, a quadratic penalty term and a Lagrange multiplier λ are introduced. The Lagrange measure is then augmented as follows.

$$l(\{\mu_k\}, \{\omega_k\}, \lambda) := \alpha \sum_k \|\partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * \mu_k(t) \right] e^{-j\omega_k t} \|_2^2 + \|f(t) - \sum_k \mu_k(t)\|_2^2 + \langle \lambda(t), f(t) - \sum_k \mu_k(t) \rangle \quad (7)$$

where α denotes the balance parameter of the data fidelity constraint. Equation (8) is then solved by the alternating direction multiplier method, the mode $\mu_k(\omega)$ in the frequency domain is updated by Eq. (8), the central frequency ω_k is updated by the equation shown in Eq. (9) and λ is updated by Eq. (10), and the $\mu_k(t)$ in the time domain is obtained from the real part of the inverse Fourier transform of $\mu_k(\omega)$ in Eq. (8).

$$\hat{\mu}_k^{n+1}(\omega) = \frac{\hat{f}(\omega) - \sum_{i \neq k} \hat{\mu}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2}, \quad (8)$$

$$\hat{\omega}_k^{n+1} = \frac{\int_0^\infty \omega |\hat{\mu}_k(\omega)|^2 d\omega}{\int_0^\infty |\hat{\mu}_k(\omega)|^2 d\omega}, \quad (9)$$

$$\hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau \left(\hat{f}(\omega) - \sum_k \hat{\mu}_k^{n+1}(\omega) \right). \quad (10)$$

VMD, as a flexible, high-resolution signal decomposition technique, has many potential applications in signal processing. It can help extract the frequency components and amplitude features in a signal, which can then be applied to tasks such as feature extraction, pattern recognition, and signal analysis. Therefore, this paper uses VMD to decompose the noise-reduced sequences to extract the feature subsequences.

Sample Entropy (SampEn) Algorithm

Richman and Moorman have developed a new metric for gauging the complexity of time series called sample entropy. The computation is more precise because sample entropy is computed as the logarithm of the total and is more closely matched with the known random fraction.

The sample entropy has two key benefits: first, it does not compare its individual data segments; instead, it is the precise value of the negative average natural logarithm of the conditional probabilities; thus it is calculated independently of the length of the data; second, it has higher consistency. This means that if one time series is valued higher than another, it is likewise valued higher for other m and r values, where r is a real number and m is an integer, representing the length of the comparison vector and “similarity” measure, respectively.

Sample entropy is a measure of sequence complexity and can be obtained by calculating the similarity of adjacent subsequences. A more regular and periodic sequence is indicated by a lower sample entropy. Therefore, the sample entropy trend can be used to determine the optimal number of VMD decomposition layers. The sample entropy of the initial few sequences falls noticeably as the number of subsequences rises because they capture the fundamental structure of the signal. However, as the subsequences increase to a certain level, the extra sequences will mainly capture the noise or the finer characteristics of the signal, which will stop the sample entropy from dramatically lowering. The number of layers of VMD decomposition can be thought of as optimal when the sample entropy value reaches a point where it plateaus or changes insignificantly, at which point the decomposed sequence can be considered to have better captured the characteristics of the original signal. Based on this observation, when the sample entropy values of the incoming nearby subsequences are close enough, we select this number of modes as the optimal number of modes.

Approximate Entropy (ApEn) Algorithm

Approximate entropy is used to describe the irregularity of a complex system, with the more irregular time series corresponding to a higher approximate entropy. The magnitude of the likelihood of a new pattern arises when the dimensionality of the reaction changes.

Specifically, approximate entropy calculates the similarity of neighboring subsequences in a sequence to assess the regularity and predictability of the sequence. In general, low-frequency sequences have higher regularity and periodicity, while high-frequency sequences are more complex and irregular. Approximate entropy can provide a quantitative assessment by calculating the entropy value of a sequence, describing the degree of irregularity of the sequence.

In this study, the $ApEn$ values of the VMD decomposed intrinsic mode function (*imf*) were calculated, and all *imf* were divided into high- and low-frequency components based on the results.

Spearman Correlation Coefficient

The Spearman correlation coefficient [40] is a non-parametric measure of the dependence of two variables. A monotonic equation is used to assess the connection between two statistical variables. If there are no repeated values in the data and the two variables are exactly monotonically connected, the Spearman correlation coefficient is either +1 or −1. Furthermore, Spearman can accurately determine the sampling probability distribution of the total without the need for a priori knowledge.

The Pearson correlation coefficient between the rank variables is what is referred to as the Spearman correlation coefficient. For a sample of sample size n , n raw data are transformed into rank data with a correlation coefficient ρ of:

$$\rho = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}, \quad (11)$$

where x, y is the corresponding element in both variables.

In practical applications, the link between the variables is irrelevant, and by subtracting the corresponding elements of the two observed variables to obtain a difference d , the above formula can also be transformed into Eq. (12).

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}. \quad (12)$$

In this study, the high- and low-frequency exogenous variables were tested by Spearman correlation coefficient and screened according to p -values.

Mutual Information (MI)

Mutual information is a special case of the broader relative entropy and is now used as a criterion for feature selection and feature transformation in machine learning, as well as for minimum redundant feature selection. In addition, mutual information has the important properties of non-negativity and symmetry. As a popular strategy for selecting features, mutual information is utilized to evaluate the significance and relevance of relationships between features and the goal variables. A higher value of mutual information indicates a higher dependency between them.

In this study, both Spearman correlation coefficient test and mutual information are used to filter the exogenous variables and filter out irrelevant variables. The joint feature filtering technique is used to test the linearity and non-linearity

of the exogenous variables to improve the efficiency of the model.

Principal Component Analysis (PCA)

The most crucial features of the data are captured by the widely used feature dimensionality reduction technique known as principal component analysis, which achieves dimensionality reduction by projecting the original data into a new feature space via a linear transformation. It transforms many intrinsically linked features into a few linearly unrelated features, and uses these few features to make a differentiated comparison of samples, and these linearly unrelated variables are called principal components. The new features not only reconstruct the original features, but also remove redundant information from the original features, making them more distinguishable. By converting the data from the original feature space to a new feature space, principal component analysis extracts the most significant features of the data. These most important features usually correspond to the direction of the main variance in the original data, i.e., the main pattern of variation in the data. Data can be reduced in dimension while retaining a large amount of information by choosing fewer major components.

To make the analysis of multi-dimensional data in this study easier and increase the computational effectiveness of the model, the principal components with a cumulative contribution of 95% were chosen.

Bidirectional Gated Recurrent Neural Network (BIGRU) Model

The GRU model is one of the most popular variants of the LSTM model and is a simplified version of the LSTM [41]. The reset gate and the update gate are the sole gates in GRU. The reset gate in the GRU controls how new input information is integrated with old memory; the smaller its value, the more information is disregarded [42]. The input gate and forgetting gate in the LSTM combine to form the update gate in the GRU. The update guys control the impact of the output hidden layer on the current hidden layer at the previous time; the larger the update gate, the greater the impact of the previous hidden layer output on the current hidden layer [43].

The output layer can access information from both the past and the future thanks to gated BIGRU, which links two hidden levels with opposing transmission directions to the same output layer. As a result, the BIGRU neural network may learn information from two independent data channels and produce predictions that are more accurate. The concept behind BIGRU is to divide conventional GRU neurons into forward and backward states (negative time direction) [44]. The reset gate and update gate are calculated as follows.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (13)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (14)$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t]) \quad (15)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \quad (16)$$

where z_t is an update gate, r_t is a reset gate, and σ is a sigmoid function. x_t denotes the input, h_t denotes the hidden state, and $\tilde{h}_t$ is a candidate vector, a modulation operation that controls the extent to which new input information is received in the cell state. W_z , W_r , and W are weight matrices.

As BIGRU considers both past and future information, it is more robust to incompleteness or uncertainty in data such noise, missing values, and outliers. It may model and speculate based on historical and future information, which improves generalization to disorganized, confusing, or incomplete data. These advantages make BIGRU a good performer and potential for applications in tasks such as time series prediction and natural language processing.

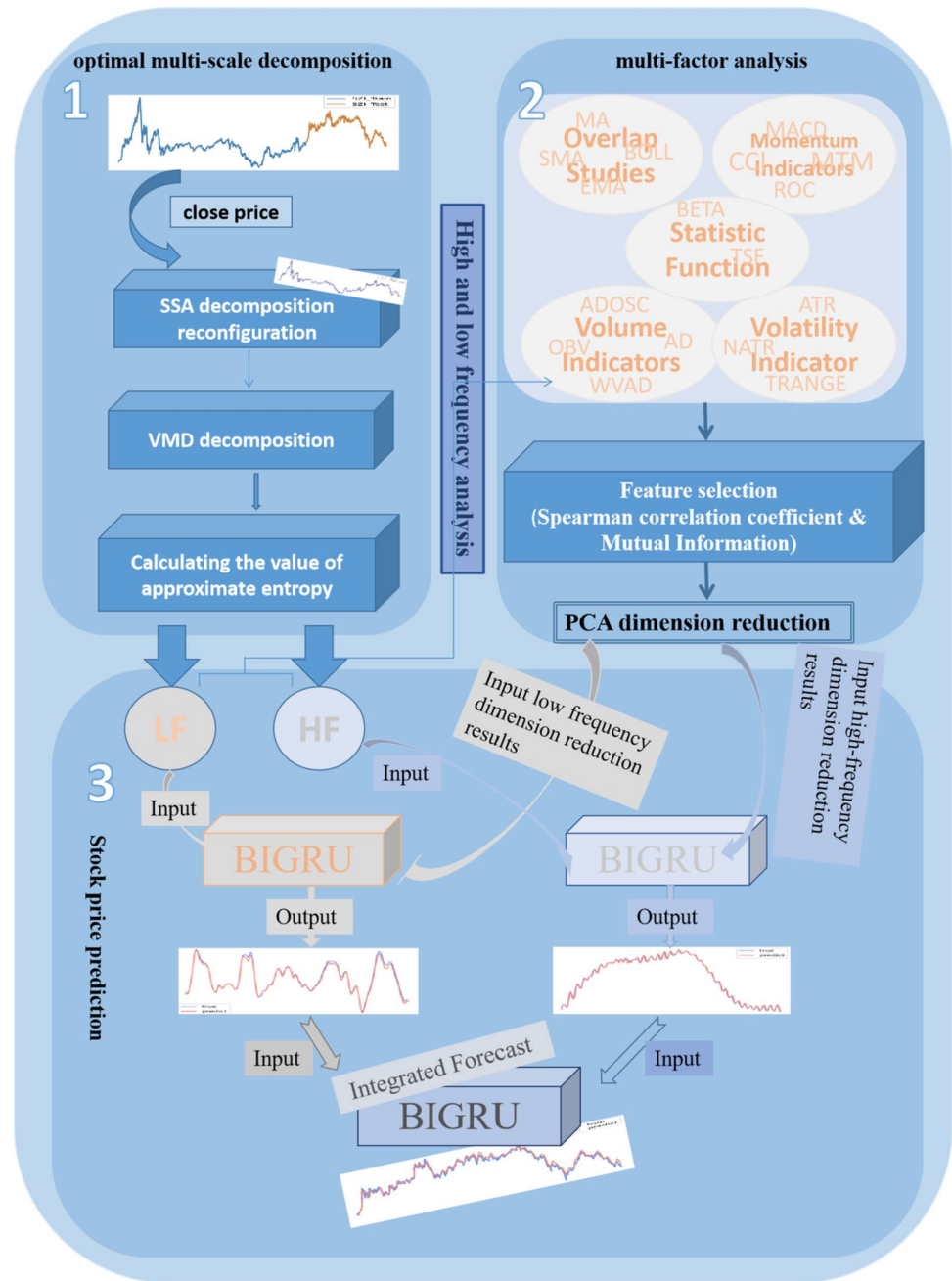
Proposed Model

Based on a two-stage decomposition integration framework, this study proposes a stock market price forecasting model, which consists of three main components, optimal multi-scale decomposition, multi-factor analysis, and forecasting, as shown in Fig. 1. The A-share market is used in this study as an example to show how well the model works. This section goes into great detail on the modeling method of the model.

Optimal Multi-scale Decomposition

This part is mainly about the extraction and reconstruction of feature subsequences. First, SSA can be used to decompose and reconstruct the stock price sequences. The noise of the reconstructed sequence is reduced to a certain extent. Then, VMD is used to decompose the de-noised complex sequence. In this step, this study abandons the method of determining the number of decomposition modes by manual setting, but uses sample entropy to scientifically determine the number of modes. Similar sample entropy values indicate that the information in two sequences is similar. In this paper, the initial mode number of sample entropy is 2. It keeps on increasing and does not change until the sample entropy values of two neighboring subsequences are sufficiently near. Finally, to distinguish between the high- and low-frequency sequences, the approximate entropy values

Fig. 1 Model framework diagram. The model framework can be divided into three parts: optimal multi-scale decomposition (in the upper left corner), multi-factor analysis (in the upper right corner), and prediction (in the lower corner). The direction of data transmission is indicated by the arrow



of the decomposition modes determined in the preceding step are calculated. Approximate entropy is a measure of time series complexity. Approximate entropy is a measure of the complexity of a time series. A higher value of approximate entropy indicates that the complexity of the sequence is higher and closer to that of a random sequence. The threshold value in this paper is the average of the approximate entropy values. When the value of the approximate entropy of a subsequence is greater than the threshold, these subsequences are considered high-frequency sequences, and similarly, low-frequency sequences are those with approximate entropy values less than the threshold. Finally, the judged

subsequences are reconfigured into a high-frequency subsequence and a low-frequency subsequence.

Algorithm

Multi-factor Analysis

In the prediction of stock prices, exogenous factors can provide more helpful data for modeling. Therefore, this article takes a more comprehensive view of the factors influencing share prices in a decomposition integration framework, that is, a multi-factor analysis of the high- and low-frequency

Input:

OS=Original sequence

1. *Subsequence= SSA decompose OS*
2. *Select the top eigentriple grouping components with large contributions and reconstruct as X*
3. **For** *i* from 2 to 9
4. *Imf_n= VMD decompose X i=decomposition the number of layers*
5. *Calculate the sample entropy*
6. *Judge if the values of the sample entropy for two adjacent subsequences are close*
7. **Return**
8. *Calculate the approximate entropy of each Imf_n*
9. *Threshold = the average of the approximate entropy values*
10. **For** *i* from 1 to *n*
11. *If the approximate entropy is greater than the threshold*
12. *Refactor these sequences into a high-frequency subsequence*
13. **Else**
14. *Refactor as a low-frequency subsequence*
15. **End**

Multi-scale decomposition based on SSA and VMD

series separately. First, the spearman correlation coefficient test and MI indicators are used to screen the low-correlation variables. The Spearman correlation coefficient test is used to compare them, and the significance level is set at 0.05, which indicates that when the p -value is less than 0.05, the linear relationship at the 5% level has statistical significance. When calculating MI, the average value of the MI value of exogenous variables is set as the threshold value. Only when the MI value of exogenous variables exceeds the threshold value, can they be considered to have non-linear correlation with the subsequence. Only exogenous variables that pass the MI index and Spearman correlation coefficient test can be kept. Second, the exogenous variables screened by the features were feature-compressed, and the principal components with a 95% cumulative contribution to the variance of the feature values were selected as a way to avoid the negative impact of high-dimensional data on modeling efficiency.

Stock Price Prediction

In making forecasts of stock prices, this paper adopts the two-stage deep integration paradigm. In this study, the two feature subsequences are modeled and predicted respectively, and the results are predicted by secondary deep

integration. First, this paper uses BIGRU to predict high and low frequency respectively. For high-frequency sequence, the input value of BIGRU is the high-frequency sequence with first-order lag and exogenous variables by feature compression, and the output result is the prediction result of high-frequency sequence. The low-frequency sequence is similar. Then, BIGRU is used for integrated prediction. The outcomes of the high- and low-frequency predictions serve as the input, and the stock closing price predictions serve as the output. In the selection of parameters, the batch size is set to 64, the activation function is set to Relu, and the loss function uses mean square error. Finally, the performance of the suggested model is compared with that of the five comparison models produced in this paper in order to confirm the validity of the model.

Empirical Study**Data Presentation**

In this paper, two major indices of the Chinese A-share market, namely the SZI (Shenzhen Stock Index) and the SSEI (Shanghai Stock Exchange), are chosen to test

the validity of the model proposed in this paper. SSEC is a sample of all stocks listed on the Shanghai Stock Exchange, reflecting the movement of stock prices on the SSEC; SZI is the main stock index of the Shenzhen Stock Exchange, reflecting the overall market movement in the SZI market. As the rise or fall of these two stocks can directly affect investor confidence, the selection of the SZI and SSEC indices in this paper is therefore illustrative and empirically valuable.

In this paper, stock data from 28 October 2014 to 28 October 2022 were selected. In each dataset, the training set accounts for 70% of the sample and the test set was represented at 30%. The time span and quantity of datasets are shown in Table 1.

Figure 2 gives the underlying trends for these stock indicators and divides the data into a training set and a test set based on modeling needs. The target for prediction is the closing index of the next day.

Table 1 Dividing the training and test sets

Dataset	Samples	Date	Number
SZI	Training set	2014.10.28–2022.06.01	1364
	Test set	2022.06.02–2022.10.28	585
SSEC	Training set	2014.10.28–2022.06.01	1364
	Test set	2022.06.02–2022.10.28	585

Input Variable Selection

When forecasting stock prices, it is important to carry out a basic analysis of not only the opening and closing prices, highs and lows, and price rises and falls of a stock, but also to consider the technical indicators of the stock. It has been established that stock technical signs are a key component of stock market modeling. Technical indicator analysis is used to study market behavioral responses and trends in stock price movements, which helps investors to time specific purchases. Therefore, in addition to the usual basic variables, this study also includes a number of representative technical indicators as influencing variables: in particular, overlap studies, MA, EMA, SMA, BOLL; momentum indicators, MTM, MACD, CCI, ROC; volume indicators, AD, ADOSC, OBV, WVAD; volatility indicator, ATR, NATR, TRANGE; and statistic function, BETA, TSF. The brief description of each technical indicator is given below.

(I) Moving average (MA).

MA is mainly used to judge the change trend of price trend. The moving average of 5 and 10 trading days is selected in this paper. The specific calculation formula is shown in formula:

$$MA_t(5) = \frac{Close_t + \dots + Close_{t-4}}{5}, \quad (17)$$

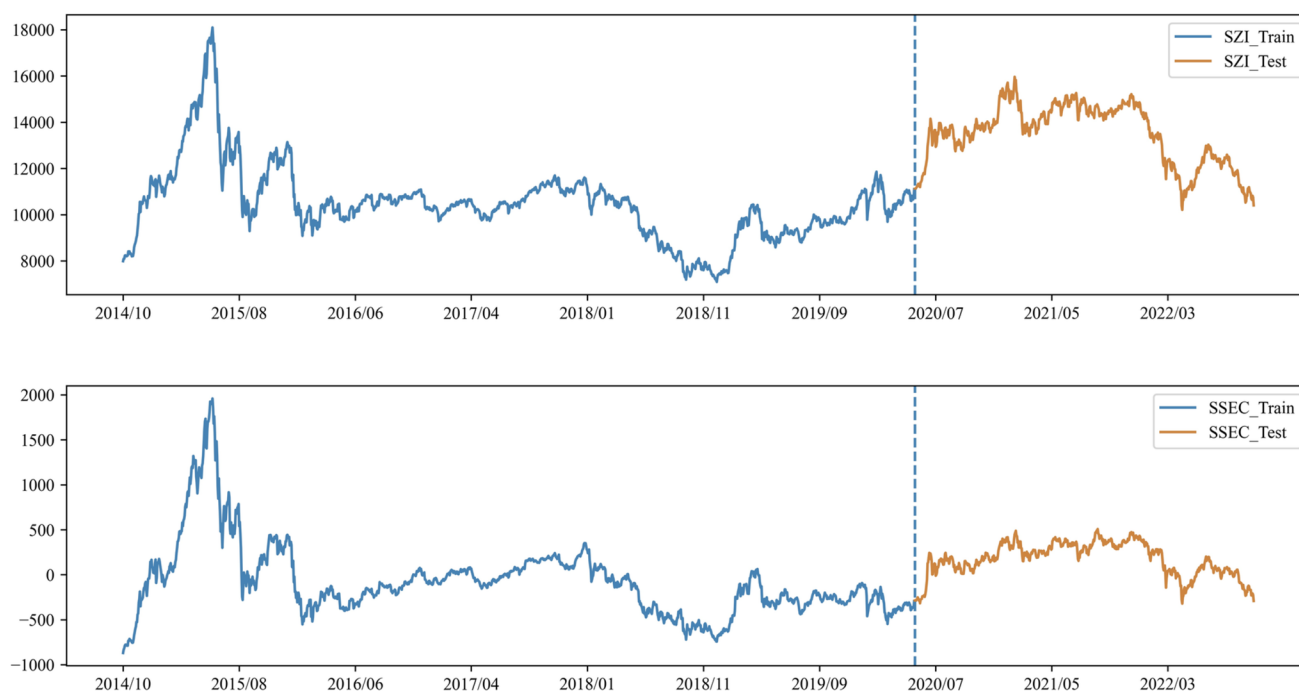


Fig. 2 Closing prices of SZI and SSEC

$$MA_t(10) = \frac{\text{Close}_t + \dots + \text{Close}_{t-9}}{10}. \quad (18)$$

(II) Simple moving average (SMA).

The SMA is calculated by dividing the total closing prices during a given period of time by that period. The more basic smoothing effect provided by the SMA may make it more effective in finding simple areas of support and resistance on a chart.

(III) Exponential moving average (EMA).

A trending indicator called the exponential moving average (EMA) uses the arithmetic average of closing price to identify the trend of upcoming price movements. The weighted influence of EMA values diminishes exponentially with time, with the weighted influence being higher for more recent data.

(IV) Moving average convergence divergence (MACD).

MACD is an indicator of asset momentum through the convergence and separation between two moving averages. It is composed of MACD line and signal line. By viewing the deviation between the MACD and the price trend, traders can gain insight into the strength of the current market trend. The formula is as follows:

$$EMA_t(12) = \frac{2}{13} \times \text{Close}_t + \frac{11}{13} \times EMA_{t-1}(12), \quad (19)$$

$$EMA_t(26) = \frac{2}{27} \times \text{Close}_t + \frac{25}{27} \times EMA_{t-1}(26), \quad (20)$$

$$DIF = EMA_t(12) - EMA_t(26), \quad (21)$$

$$DEA = DIF_t \times \frac{2}{10} + DEA_{t-1} \times \frac{8}{10}, \quad (22)$$

$$MACD = (DIF - DEA) \times 2. \quad (23)$$

(V) Commodity channel index (CCI).

The primary purpose of CCI is to determine whether the stock price has risen above the normal distribution range. It is mainly applicable to overbought and oversold areas, and can more accurately detect the sharp rise and fall of the market. It is very suitable for short-term operations in stock, foreign exchange, and other markets. The formula is as follows:

$$CCI(12) = \frac{TP - MA(12)}{0.015 \times MD(12)}, \quad (24)$$

$$TP(12) = \frac{\text{High}_{12} + \text{Low}_{12} + \text{Close}}{3}. \quad (25)$$

(VI) Average true range (ATR).

ATR is the trading fluctuation after averaging the N day index moving average. The formula is as follows:

$$A = \text{High}_t - \text{Low}_t, \quad (26)$$

$$B = \text{Close}_{t-1} - \text{High}_t, \quad (27)$$

$$C = \text{Close}_{t-1} - \text{Low}_t, \quad (28)$$

$$TR = \max\{A, B, C\}, \quad (29)$$

$$ATR(14) = \frac{TR_t + \dots + TR_{t-13}}{14}. \quad (30)$$

(VII) Normalized average true range (NATR).

With normalized ATR, attention needs to be paid to the range after normalization.

(VIII) True range (TRANGE).

The true volatility is the maximum of the volatility between the highest and lowest prices of the previous trading day, the volatility between the closing price of the previous trading day and the highest price of the current trading day, and the volatility between the closing price of the previous trading day and the lowest price of the current trading day.

$$\begin{aligned} \text{TrueHigh} &= \text{Highestofhigh}_{[0]} \text{or close}_{[-1]} \\ \text{TrueLow} &= \text{Lowestoflow}_{[0]} \text{or close}_{[-1]} \\ TR &= \text{TrueHigh} - \text{TrueLow} \end{aligned} \quad (31)$$

(IX) Momentum index (MTM).

MTM is the use of the principle of constant velocity buffer to observe the speed of rise and fall in share prices, to analyze the speed of share price fluctuations, and to study the various accelerating and decelerating inertial effects of share prices in the course of fluctuations. Generally, as the share price rises, so does the momentum value. In the absence of other non-technical influences, the MTM broadly reflects the speed of share price change.

$$MTM(n) = \text{Close}_t - \text{Close}_{t-n}. \quad (32)$$

(X) Bollinger bands (BOLL).

In order to determine the range of fluctuations and future trends of the stock price, the BOLL applies statistical principles to calculate the standard deviation and confidence interval of the stock

price. The BOLL is then used to show the safe high and low prices of the share price. When the Bollinger bands become wider from narrow, it means that the stock price volatility increases and trading is active. If the stock price approaches or crosses the Bollinger Bands upwards, it indicates that overbought forces are increasing and the stock market may fall in the short term. Conversely, if the stock price approaches or crosses the Bollinger Bands downwards, it indicates that oversold forces are increasing and the stock market may rebound in the short term.

(XI) Rate of change (ROC).

By contrasting the stock price on 1 day with the stock price on the day before, ROC illustrates the rate of change of the stock market. The formula is as follows:

$$ROC(12) = \frac{Close_t - Close_{t-12}}{Close_{t-12}}. \quad (33)$$

(XII) William's variable accumulation distribution (WVAD).

WVAD is a volume indicator that measures the strength of both the long and short sides of a share price from the opening to the closing of the market. The main theoretical essence of WVAD is to focus on the price levels between the opening and closing of the day, to consider the price levels above this area as pressure and those below the area as support, and to find the percentage of the total fluctuation of the area for the day, in order to measure how much of the day's volume belongs to this area. This is used to measure how much of the day's volume falls within this zone and becomes actual meaningful trading volume.

(XIII) Chaikin A/D line (AD).

This is an equilibrium volume indicator proposed by Marc Chaikin, which uses the closing price of the day to measure trading volume and is used to measure the total amount of money accumulated in a security over a period of time.

$$Longandshortcontrast = \frac{(Close - Low) - (High - Close)}{(High - Low)}. \quad (34)$$

(XIV) Chaikin A/D oscillator (ADOSC).

ADOSC detects inflows and outflows of funds in the market by comparing capital flows with price behavior. Bullish divergences go long, bearish divergences go short; positive to negative sell, negative to positive buy.

$$ADOSC = \text{fastperiodA/D} - \text{slowperiodA/D}. \quad (35)$$

(XV) On-balance volume (OBV).

OBV speculates the stock price change trend by calculating the change trend of trading volume. Based on a certain day, the total trading volume of listed stocks is accumulated every day. If the index or stock rises every other day, the benchmark period OBV plus today's trading volume is today's OBV. If the index or stock falls the next day, the benchmark period OBV minus the trading volume of the day is the OBV of the day. The formula is as follows:

$$OBV = \frac{(Close - Low) - (High - Close)}{(High - Low) \times vol}. \quad (36)$$

(XVI) BETA

BETA is used to gauge how volatile individual stocks or equity funds' prices are in comparison to the entire stock market. Beta is a relative indicator that assesses the overall volatility of stock returns compared to the return of its performance benchmark. The higher β , the higher the volatility. If β exceeds 1, the volatility of the stock exceeds the volatility of the performance benchmark. The opposite is also accurate.

(XVII) Time series forecast (TSF).

TSF is an extension of historical data forecasting. It is a technique for predicting the development trends of socio-economic phenomena by extrapolating the development process and regularity as seen in the time series.

Evaluation Criterion

Three statistical indicators, MAE, RMSE, and MAPE, were used for this study to qualitatively assess the prediction effectiveness of all pertinent models (Table 2). Generally speaking, the better the accuracy of the model, the smaller the value of the indicator.

Table 2 Evaluation criterion

Evaluation criterion	Formula
Root mean square error (RMSE)	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$
Mean absolute error (MAE)	$MAE = \frac{1}{n} \sum_{i=1}^n Y_i - \hat{Y}_i $
Mean absolute percentage error (MAPE)	$MAPE = \frac{1}{n} \sum_{i=1}^n \left \frac{Y_i - \hat{Y}_i}{Y_i} \right $

Optimal Multi-scale Decomposition

In this section, the optimal multi-scale decomposition is used for feature extraction and reconstruction of the optimal subsequence for SSEC and SZI, and the specific construction process is shown in Figs. 3 and 4.

In this study, SSA was first used to decompose the stock closing prices. The figure shows the top ten component series sorted by singular value after decomposition. It can be seen that in all stock indices except the first 3 series, the remaining ones can basically be regarded as noisy series. Therefore, the first 3 series are reconstructed. Compared to the original series, the reconstructed series are significantly smoother, but at the same time maintain the overall variability.

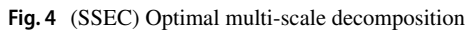
Next, the reconstructed sequences are decomposed in two stages using VMD. In this step, this paper discards the method of determining the number of decomposition modes by manual settings and instead uses sample entropy to scientifically determine the number of modes. Similar sample

entropy values indicate equivalent information content between two sequences. The initial mode number of sample entropy in this paper is 2. It keeps increasing and not to change until the sample entropy values of the two neighboring subsequences are sufficiently near to one another. In Fig. 3, the sample entropy values for each component obtained by decomposing at various modal numbers are displayed, with the final modal number selected denoted in bold black. In the case of SZI, for example, when the modal number is 8, there is information redundancy in the last two components; the modal number is therefore set to 7. Similarly, it is established that the decomposition level of the SSEC data is determined to be 8. The final decomposition results are shown in the figure, where $imf1, imf2, \dots, imfn$ denote the respective components obtained by VMD, and n denotes the number of decomposition levels.

Since each subsequence acquired following VMD decomposition carries a variable amount of information, pseudo-components may sometimes arise. In this paper, the approximate entropy values of each sub-series after



Fig. 3 (SZI) Optimal multi-scale decomposition



Multivariate Analysis

In this paper, the Spearman correlation coefficient test and the MI indicator were used to collaboratively extract exogenous variables. The relationships between the variables and the high- and low-frequency series are shown in Figs. 5 and 6. When performing the Spearman correlation coefficient test, when the p -value is less than 0.05, it indicates a statistically significant linear relationship at the 5% level. When calculating MI, exogenous variables were considered to be non-linearly correlated with the subsequence only if their MI values exceeded the mean. Therefore, only exogenous variables that passed the Spearman correlation coefficient test and the MI index were retained. As shown in Figs. 5 and 6, the SZI high-frequency series and SSEC low-frequency series

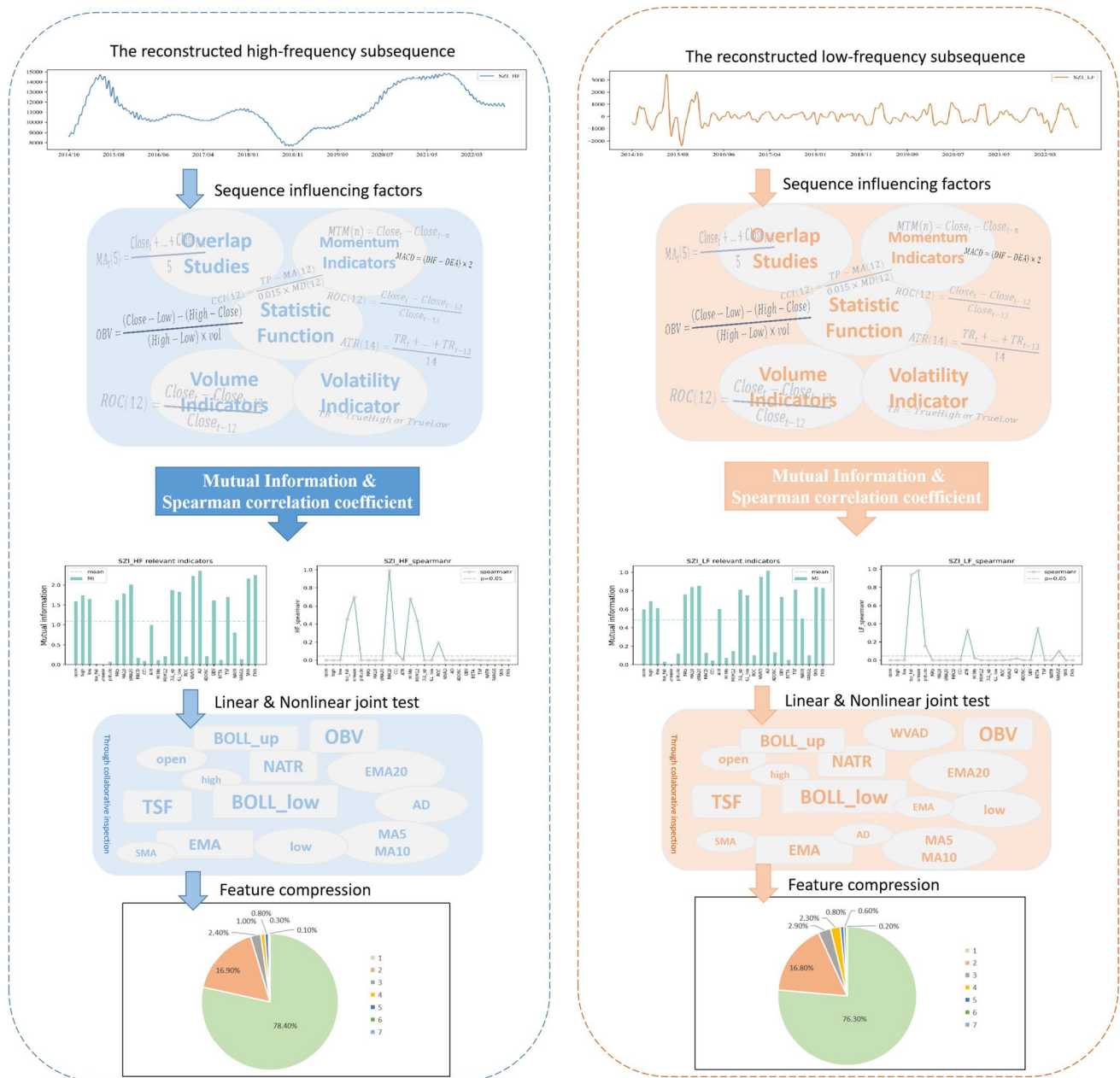


Fig. 5 (SZI) Multivariate analysis

were retained by screening for 14 variables, namely open, high, low, MA5, MA10, EMA20, BOLL_up, BOLL_low, WVAD, AD, OBV, TSF, SMA, and EMA. The SZI low-frequency series were retained for open, high, and low. The SSEC high-frequency series retains 10 variables, namely open, high, MA5, MA10, EMA20, BOLL_low, TSF, NATR, EMA20, and EMA. The variables that passed the filter were subsequently included in the model for further analysis.

Considering the high dimensionality of the filtered variables and the impact on the efficiency of the modeling, therefore, this study uses PCA for feature compression. The

compressed new features not only reconstruct the original features, but also remove the redundant information from the original features, making them more distinguishable. This method allows the data to be compressed without reducing the effective information, thus achieving a reduction in dimensionality. The figure shows the percentage of variance of the principal components of each sequence after PCA compression. In this paper, the principal components with a cumulative contribution of 95% of the variance of the eigenvalues are selected as a way to prevent the implications of high-dimensional data on modeling effectiveness. Take

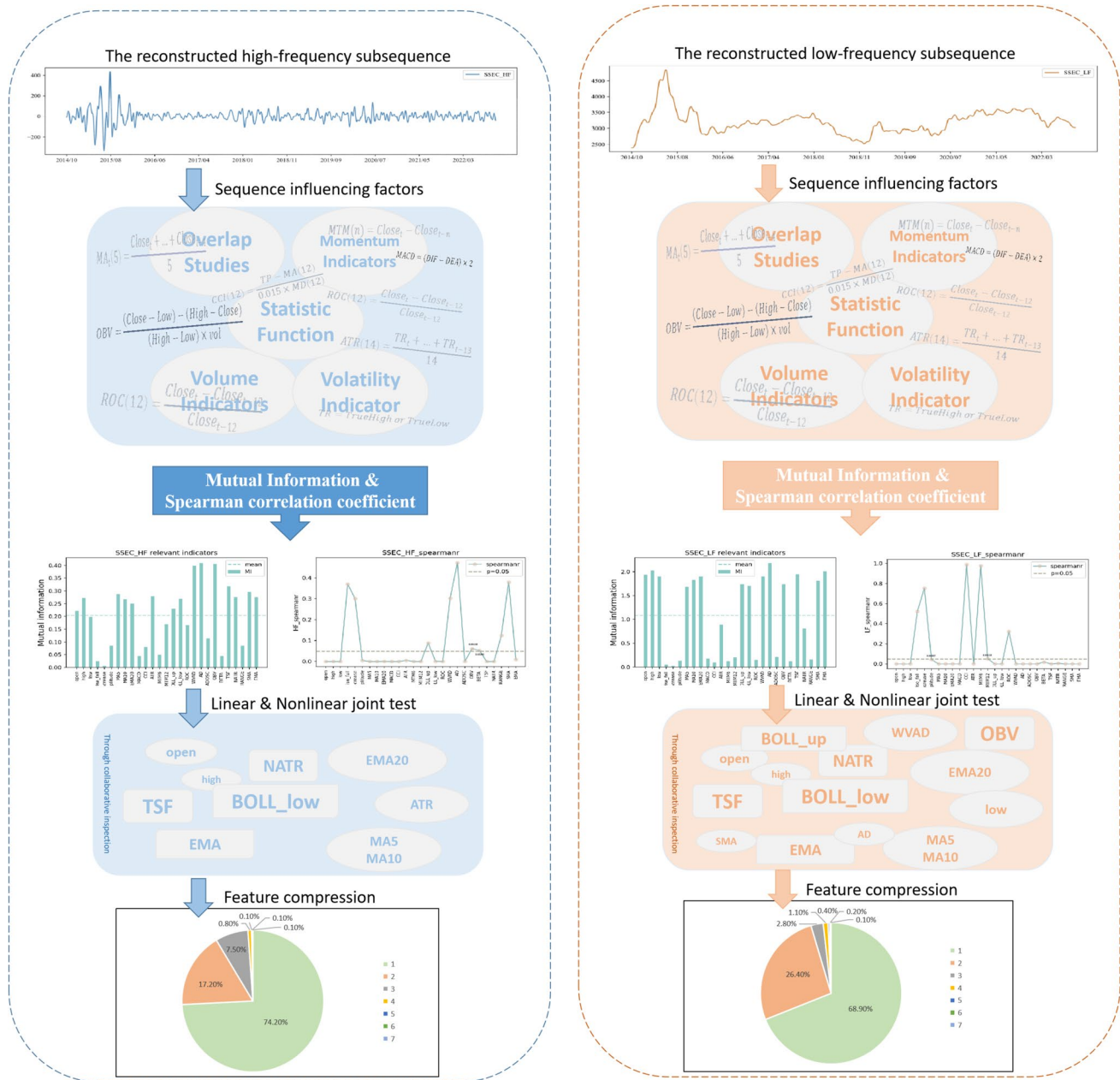


Fig. 6 (SSEC) Multivariate analysis

SZI as an example, as shown in Fig. 5; when the dimension is 2, the cumulative variance contribution of SZI high-frequency sequences reaches 95.3%; therefore, SZI high-frequency sequences are compressed to 2 dimensions; when the dimension is 3, the cumulative variance contribution of SZI low-frequency sequences reaches 96%; therefore, SZI low-frequency sequences are compressed to 3 dimensions. Similarly, SSEC high-frequency sequences were compressed to 3 dimensions and SSEC low-frequency sequences were compressed to 2 dimensions.

Analysis of Integrated Prediction Result

In the prediction stage, BIGRU is used in this paper, which can better extract the bidirectional dependency in the time series and give higher weight to the filtered high value information. The input layer for data, the BIGRU, and the output layer make up the majority of the prediction model. In this paper, high- and low-frequency data are predicted respectively, and the input data are normalized. Then, the high- and low-frequency forecast results are used as the input of the final integrated forecast to get the final stock

closing price forecast results. All the predicted results are shown in Fig. 7.

For high-frequency series, BIGRU takes the high-frequency series with a one order lag and the reconstructed exogenous variables as inputs, and it outputs the expected result for the high-frequency sequence. Using the SZI market as an example, the subsequence in Fig. 3 is divided by the approximate entropy, and the portion above the mean is considered a high-frequency series and reconstructed and entered into the model. The exogenous variables of the high-frequency series are shown in Fig. 5. A total of 14 variables, open, high, low, MA5, MA10, EMA20, BOLL_up, BOLL_low, WVAD, AD, OBV, TSF, SMA, and EMA, are retained by Spearman's correlation coefficient and joint test. They are then compressed by features into two dimensions

and entered into BIGRU. The end prediction of the high-frequency series is displayed on the graph in the top left corner of Fig. 7. The red line displays the predicted values, while the blue line displays the actual values. It can be seen that the combination model produces accurate predictions.

Similarly, for the prediction of low-frequency series, the input value of BIGRU is the low-frequency series with one order lag and the reconstructed exogenous variable, and its output is the prediction result of the low-frequency series. Using the SZI market as an example, the subsequences in Fig. 3 are divided by the approximate entropy, and the part below the mean is considered low-frequency series and reconstructed and entered into the model. Figure 5 shows the exogenous variables of the low-frequency series. A total of 15 series of open, high, low, MA5, MA10, EMA20,

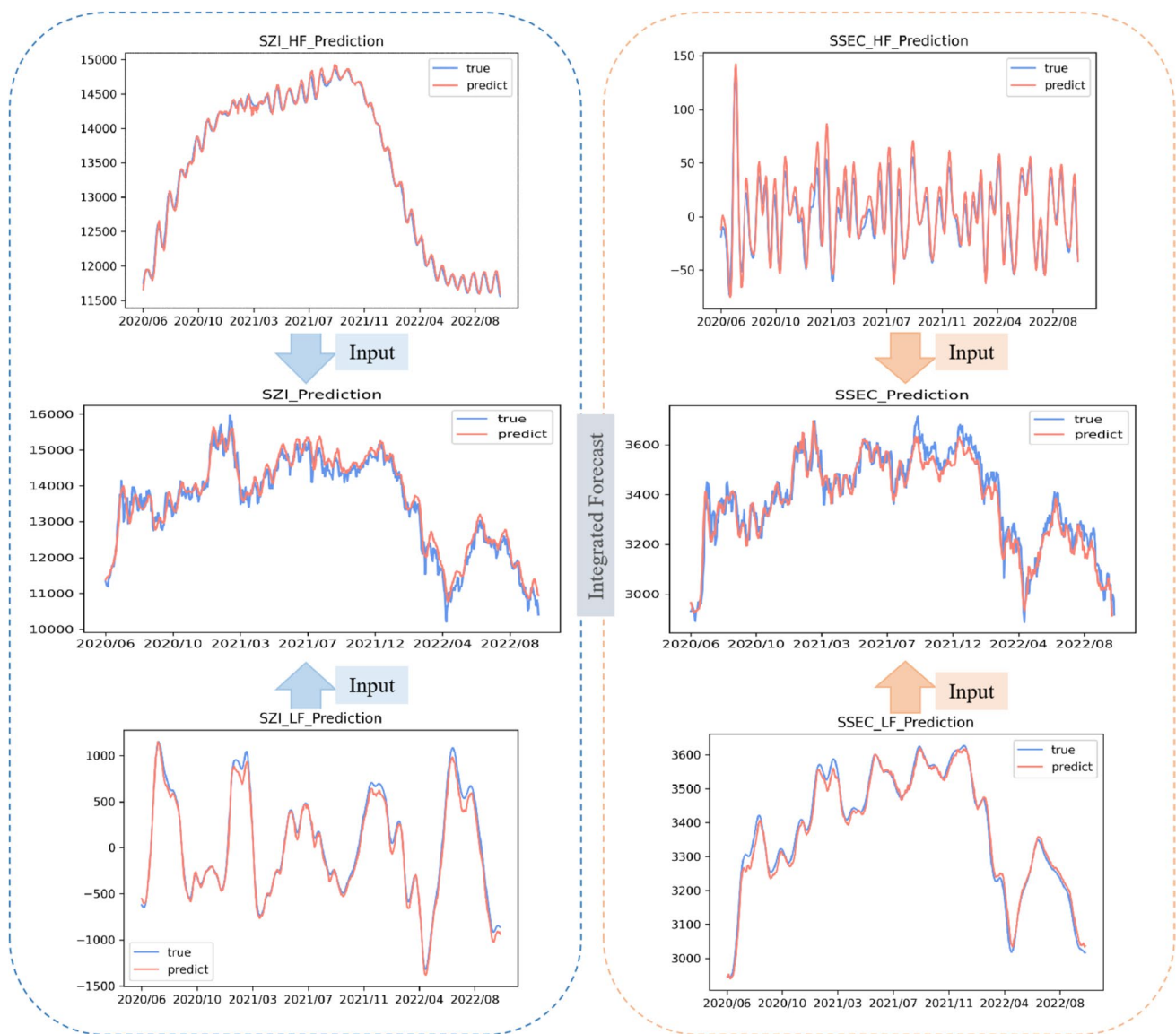


Fig. 7 All predicted results

BOLL_UP, BOLL_LOW, WVAD, AD, OBV, TSF, NATR, SMA, and EMA were retained by a joint Spearman correlation coefficient and MI test. They were then compressed into three dimensions by features and fed into BIGRU. The final predictions for the low-frequency sequences are shown in the bottom left of Fig. 7. The red line displays the predicted values, while the blue line displays the real values. It is clear that the combination model produces accurate predictions.

Finally, the integrated prediction is performed with BIGRU, with the input being the high- and low-frequency predictions and the output being the final prediction of the stock closing price. The final prediction findings of the model are displayed in the middle of Fig. 7. The real value is shown by the blue line, while the predicted value is shown by the red line. It has been discovered that the model produces accurate forecasts.

As can be seen from Table 3, the integrated prediction method based on multi-scale decomposition and multi-factor analysis (SSA-VMD-MFA-BIGRU) proposed in this paper is optimal in all cases, with the lowest values of the error evaluation indexes in the whole process, which are 171.1673, 221.2086, and 1.2796% on the SZI and 29.9973, 39.3508, and 0.8946% on the SSEC, respectively.

Analysis and Discussion

In order to verify the validity and stability of the decomposition integration prediction framework proposed in this paper, seven comparative models are built based on excellent papers in related research areas. These models are the most

advanced research results in recent years and are all typical. The comparison results of the two stock indices are shown in Table 3 and Fig. 8.

Comparison with Prediction Models

This section compares the model proposed in this paper with GRU [45] and CNN-BILSTM [47]. In order to show the performance of each model more visually, the bar chart shown in Fig. 8 is plotted in this paper.

From Fig. 8 and Table 3, it can be seen that among all the models, the model proposed in this paper has the best evaluation metrics. The performance of the single prediction model GRU on the three evaluation indicators is 221.3521, 290.9207, and 1.6643% respectively, which is still some way off from our proposed model, saying that the single prediction model is not effective enough in prediction. In addition to single prediction models, hybrid prediction models combining multiple prediction models, such as CNN-BILSTM, are gradually gaining popularity. However, from the experimental results, the prediction performance of such models is even worse than that of single prediction models, indicating that such models are not stable and have a large room for optimization. Therefore, if the stock price data is not processed and directly put into the model for prediction, it will have some influence on the final prediction results and reduce the accuracy of the prediction.

Comparing SSA-SVR [38] with SSA-BIGRU [48], we can find that SSA-BIGRU outperforms SSA-SVR by 187.7221, 247.5064, and 1.4110% on the three evaluation indicators on the SZI and 76.6564, 89.0099, and 2.2300% on the SSEC, respectively. Therefore, we can see that the deep learning model outperforms the traditional machine learning model in terms of stock price prediction. Furthermore, in the case of the SZI market, SSA-BIGRU outperformed CNN-BILSTM and GRU in terms of evaluation metrics, so the decomposition-based deep learning model also outperformed the single prediction model.

Comparison with Multi-scale Decomposition Models

The above comparative experiments revealed that we need to process the share price when forecasting. By decomposing the stock price, we can get a more comprehensive understanding of the trend, noise, etc. of the stock price, and thus make better investment and trading decisions. Therefore, the proposed model is compared with CEEMDAN-LSTM [46], SSA-SVR [38], SSA-BIGRU [48], VMD-LSTM [22], and VMD-SE-GRU [49] in this paper. The specific results are shown in Table 3 and Fig. 8.

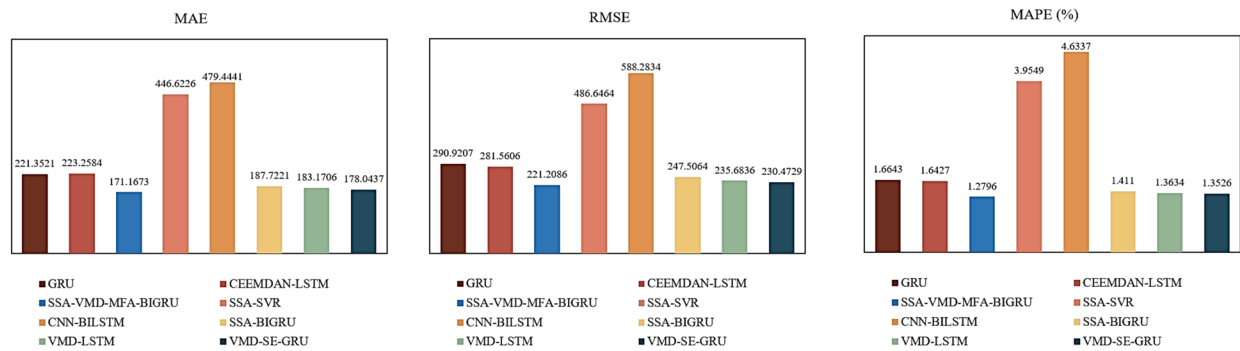
In the SZI, all the prediction models based on multi-scale decomposition outperformed the direct prediction, except SSA-SVR. The results show that most of the prediction

Table 3 Comparison of the empirical results of the model and the proposed model

Market	Model	MAE	RMSE	MAPE (%)
SZI	GRU [45]	221.3521	290.9207	1.6643
	CEEMDAN-LSTM [46]	223.2584	281.5606	1.6427
	SSA-VMD-MFA-BIGRU	171.1673	221.2086	1.2796
	SSA-SVR [38]	446.6226	486.6464	3.9549
	CNN-BILSTM [47]	479.4441	588.2834	4.6337
	SSA-BIGRU [48]	187.7221	247.5064	1.4110
	VMD-LSTM [22]	183.1706	235.6836	1.3634
	VMD-SE-GRU [49]	178.0437	230.4729	1.3526
SSEC	GRU [45]	37.3347	50.9810	1.1155
	CEEMDAN-LSTM [46]	102.4273	123.4081	3.0580
	SSA-VMD-MFA-BIGRU	29.9973	39.3508	0.8946
	SSA-SVR [38]	109.1372	117.2263	3.3783
	CNN-BILSTM [47]	133.8181	167.2553	4.5655
	SSA-BIGRU [48]	76.6564	89.0099	2.2300
	VMD-LSTM [22]	32.7635	41.5849	0.9680
	VMD-SE-GRU [49]	30.5862	39.5800	0.9073

1

(SZI) Error evaluation indicators



2

(SSEC) Error evaluation indicators

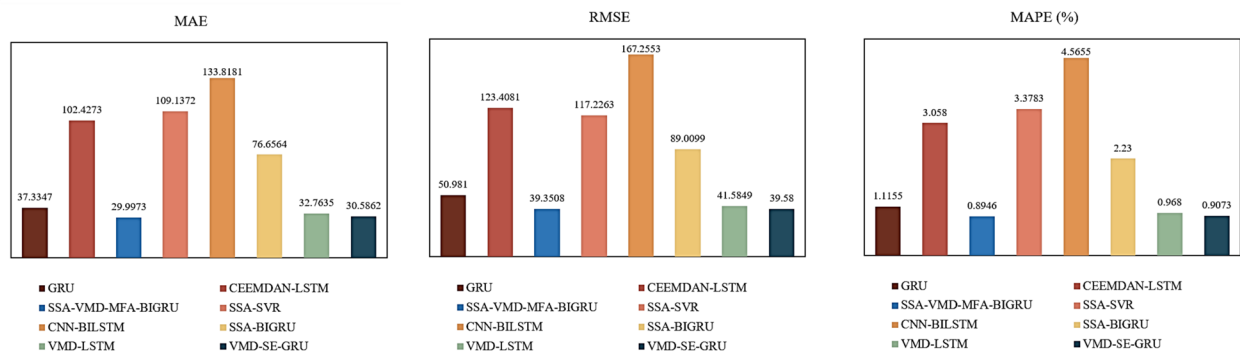


Fig. 8 Error evaluation indicator values

models based on decomposition outperformed the direct prediction; therefore, the signal decomposition of the original data is helpful to improve the prediction accuracy of the model.

A comparison of SSA-SVR, SSA-BIGRU, VMD-LSTM, and VMD-SE-GRU reveals that different decomposition methods also have an impact on the prediction results. Regardless of SZI or SSEC markets, the VMD-LSTM and VMD-SE-GRU models outperformed SSA-SVR and SSA-BIGRU on all three evaluation indicators, which to some extent indicates that the decomposition using VMD is better. In addition, the performance of VMD-SE-GRU on the three evaluation indicators was 178.0437, 230.4729, and 1.3526% on the SZI and 30.5862, 39.5800, and 0.9073% on the SSEC, respectively, all of which were better than VMD-LSTM, indicating that using a scientific method to determine the optimal number of decomposition layers of VMD can also improve the accuracy of prediction to a certain extent. This indicates that the optimal number of decomposition layers of VMD can also improve the prediction accuracy of the model to a certain extent.

The SSA-VMD-MFA-BIGRU model proposed in this paper was 171.1673, 221.2086, and 1.2796% on the SZI and 29.9973, 39.3508, and 0.8946% on the SSEC, respectively.

The model uses a quadratic decomposition, which gives a significant improvement in prediction compared to the single decomposition model described above, suggesting that the single decomposition is not as effective as the quadratic decomposition model. At the same time, it can also be seen from the comparison model that adding multi-factor analysis can provide more useful data for the development of the model and enhance its predictive power.

Summary of Comparative Experimental Results

As shown in Table 3 and Fig. 8, the combined SSA-VMD-MFA-BIGRU model proposed in this paper performs significantly better than the other comparative models in MAE, RMSE, and MAPE in both markets, and its results also indicate that the model has a more accurate and stable forecasting performance. The specific experiments are summarized below.

- (1) Individual prediction methods are the basis of research, but successful outcomes can only be obtained by ongoing development and fusion with other superior techniques. The above experiments have also indirectly

demonstrated that it is difficult for a single prediction model to make accurate predictions.

- (2) Decomposition techniques are the key to improving prediction performance. The prediction results based on the quadratic decomposition method outperformed the single decomposition and direct prediction. This shows that the quadratic decomposition technique proposed in this paper is effective.
- (3) Multi-factor analysis is an important way to improve the prediction performance. Comparative experiments show that the inclusion of multi-factor analysis can effectively improve the model performance, which also indicates that it is necessary to include multi-factor analysis in this paper.
- (4) The model proposed in this paper performs best in all comparative experiments, which indicates that the model has good adaptability in stock price forecasting. It not only proposes a hybrid forecasting method in a decomposition integration framework, but also shows that the innovative combination strategy outperforms the single model. The effectiveness of the proposed model is also well demonstrated by the minimum error evaluation metric.

Conclusions

Stocks play an important role in the financial markets. To more effectively assist financial institutions in helping investors deliver financial early warning and help investors avoid losses in the financial markets, stock price forecasting has emerged as a pressing issue that requires attention today. In this paper, a two-stage deep integration paradigm based on optimal multi-scale decomposition and multi-factor analysis is proposed, which includes an optimal decomposition algorithm for extracting and reconstructing feature subsequences, multi-factor analysis, and a two-stage deep integration paradigm. These methods are used to analyze the characteristics of stock prices, providing support for the forecasting method proposed in this paper in terms of plausibility and scientific validity. Also, this study verifies the effectiveness of this model by comparing it with seven comparative models in two markets. Through experimental analysis and discussion, the proposed prediction model has the following advantages:

Firstly, in terms of capturing hidden information from the original data, the optimal multi-scale decomposition method performs better than the single decomposition method. At the same time, multi-scale decomposition aids in reducing noise interference, enhancing the accuracy and stability of the model, and capturing trends and variations at many time scales, offering a more thorough analytical perspective. Secondly, multi-factor analysis can significantly improve the predictive power of a model and provide more

relevant data for model development. Thirdly, the two-stage deep integration paradigm is superior to the simple linear accumulation of subsequences in improving the prediction accuracy of the model. Compared to linear accumulation, non-linear integrated prediction improves generalization to data and reduces the risk of overfitting. Fourth, compared with other prediction models, the prediction model based on the decomposition integration framework proposed in this study greatly improves the prediction accuracy.

This study also has some limitations. First, the stock market is deeply affected by factors such as investor sentiment, and in future research, quantitative studies on investor sentiment and other factors can be conducted to comprehensively analyze the influencing factors of stock prices in order to improve the prediction accuracy of the model. Second, although this paper proposes a decomposition integration framework based on optimal multi-scale decomposition and multi-factor analysis, with the further development of machine learning, more advanced models can be selected to improve the decomposition integration framework in the future in order to further improve the prediction accuracy. Third, although point prediction can improve the model prediction accuracy, future research can consider interval prediction on this basis to provide a more effective reference basis for investment decisions.

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Data Availability The data that support the findings of this study are sourced from 10.jqka.com.cn.

Declarations

Ethical Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed Consent Informed consent was obtained from all individual participants included in the study.

Conflict of Interest The authors declare no competing interests.

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